$y_i = wx_i + b$

y = single continuous output value

w = weight / slope of line (list of weights)

x = list of input values.

$$L(w,b) = \frac{\sum_{i=0}^{n}\left[\left(y_{i_t} - (wx_i+b)\right)^2\right]}{N}$$

(MSE ERROR)

N = number of input points

y_{i_t} = Actual Output

$wx_i + b$ = predicted output

$$\frac{\partial L}{\partial w} = \frac{2}{N} \times \sum_{i=0}^{n}\left(y_{i_t} - (wx_i+b)\right) \times (-x_i)$$

$L(w,b)$ = Loss function

$$\frac{\partial L}{\partial w} = \frac{-2}{N}\left[\sum_{i=0}^{n}\left(y_{i_t} - (wx_i+b)\right)\right] \times (x_i)$$

$$\frac{\partial L}{\partial b} = \frac{2}{N} \times \sum_{i=0}^{n}\left(y_{i_t} - (wx_i+b)\right) \times (-1)$$

$$\frac{\partial L}{\partial b} = \frac{-2}{N}\left[\sum_{i=0}^{n}\left(y_{i_t} - (wx_i+b)\right)\right]$$

Gradients of Loss function. (Partial Derivatives)

↓

Denotes Rate of change of Loss function and direction of change.

(− if minimizing)

Learning Rate

$$w_{new} = w_{old} - \eta \cdot \frac{\partial L}{\partial w}$$

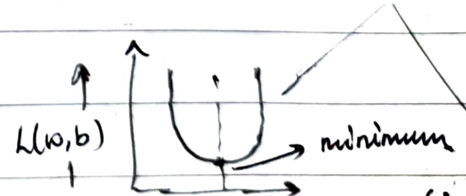
negative because

is $\frac{\partial L}{\partial w} \downarrow (-)$, we should add weights $- \times (-) = +$

if $\frac{\partial L}{\partial w} \uparrow (+)$, we should remove weights $- \times (+) = -$

$\eta \uparrow$ - Shoots beyond minima, oscillates wildly

$\eta \downarrow$ - Takes forever to reach minima



$\frac{\partial L}{\partial w}$ is − (increase weight) Loss decreasing

$\frac{\partial L}{\partial w} = 0$ (perfect)

$\frac{\partial L}{\partial w}$ is + (L in… (overshoot decrease weight)